

ET5218 INNOVATION AND COMMERCIALIZATION OF TECHNOLOGIES

05 New Service Development

NTU Innovation and
Entrepreneurship

Classification: Restricted



Learning Outcomes

1. Examine the characteristics of Service and Service Innovation.
2. Understand the importance of Public Service Innovation and Public Value

Growth in Services

The main change in the development of the economies over the past 20 years has been the growth of knowledge-based economy.

2021 - The global average of value-added services as a percentage of the GDP stands at 55.43%.

Countries ▾ ▾	Share of services, 2021 ▾ ▾	Global rank ▾ ▾	Available data ▾ ▾
Macao	94.18	1	1991 - 2021
Lebanon	94.15	2	1994 - 2021
Bermuda	90.5	3	2010 - 2021
Hong Kong	89.59	4	2000 - 2021
Monaco	80.09	5	2010 - 2021
Luxembourg	79.32	6	1995 - 2021
Bahamas	78.58	7	1984 - 2021
Malta	77.69	8	1970 - 2021
USA	77.6	9	1997 - 2021
Maldives	73.25	10	2003 - 2021
Palau	73.23	11	2000 - 2021
Cyprus	73.22	12	1995 - 2021
Israel	72.45	13	1995 - 2021
S.T.&Principe	72.25	14	2001 - 2021
Saint Lucia	72.16	15	2006 - 2021
Switzerland	71.92	16	1990 - 2021
UK	71.46	17	1990 - 2021
Barbados	71.1	18	1975 - 2021
France	70.34	19	1960 - 2021
Samoa	69.79	20	1999 - 2021
Singapore	69.45	21	1960 - 2021

Source: TheGlobalEconomy.com (2023). *Share of Services by country, around the world*. https://www.theglobaleconomy.com/rankings/share_of_services/ accessed 10 May 2023

Services in Singapore

Sectors with a big net increase in businesses in 2022

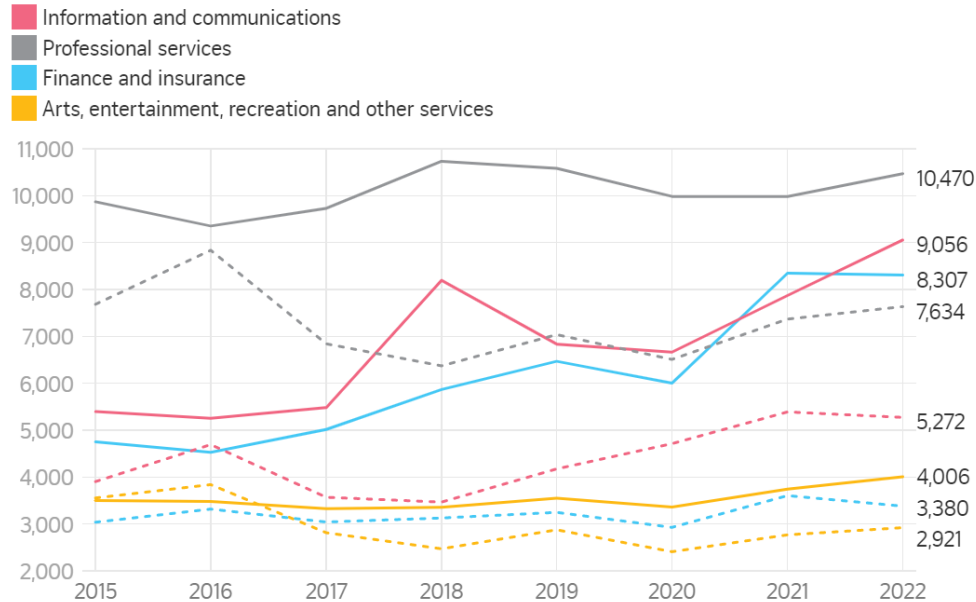


Chart: STRAITS TIMES GRAPHICS • Source: ACCOUNTING AND CORPORATE REGULATORY AUTHORITY (ACRA)

Source: Retrieved from <https://www.straitstimes.com/business/64000-new-business-formations-and-50000-business-closures-in-2022> on 29 Jan 2023

Services in Singapore

Speech | 28 FEB 2023

Speech by Minister for Trade and Industry Gan Kim Yong at Ministry of Trade and Industry (MTI)'s Committee of Supply Debate 2023

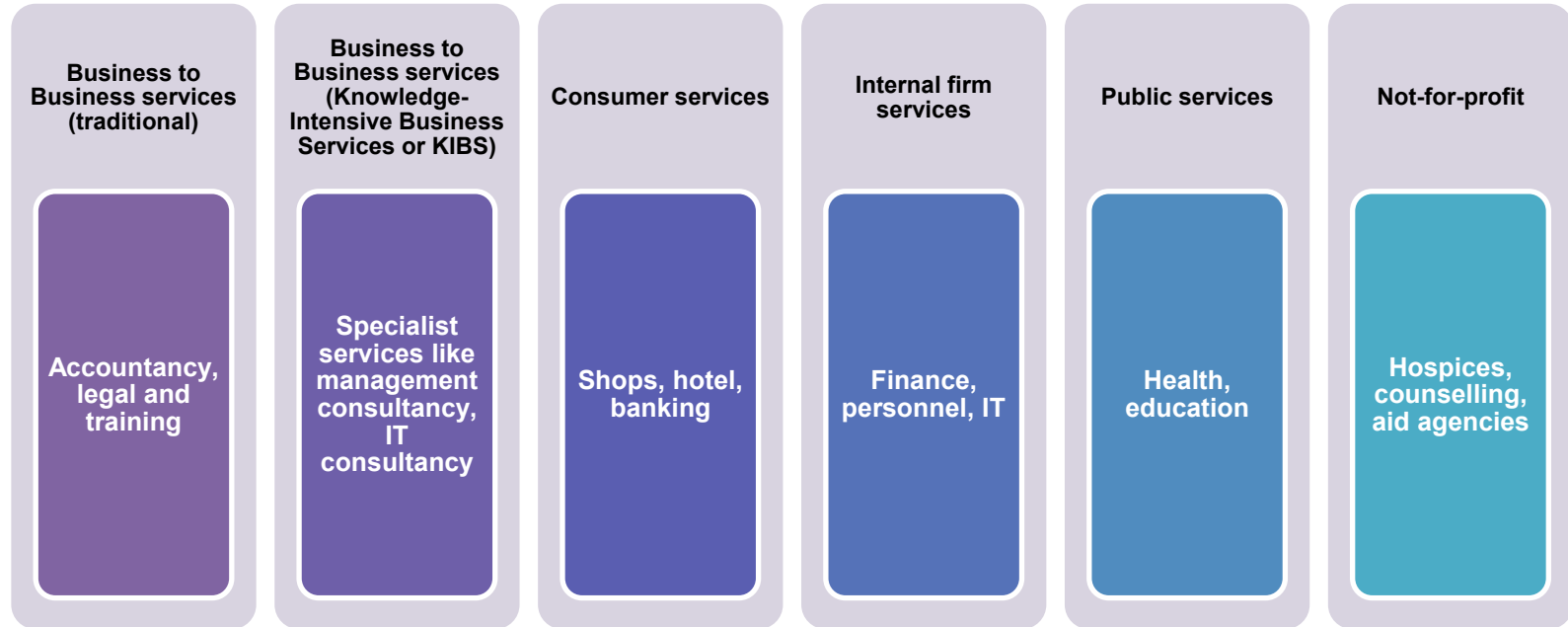
"Building a Vibrant Economy, Nurturing Enterprises"

2016 – 2021: Net Creation 134,000 jobs for Professional Services (e.g. Company HQ, consulting, legal & accounting), financial services, infocomm tech

2030: 100,000 Jobs for Singapore Economy 2030 Vision

Source: <https://www.mti.gov.sg/Newsroom/Speeches/2023/02/Speech-by-Minister-for-Trade-and-Industry-Gan-Kim-Yong-at-Ministry-of-Trade-and-Industry> accessed on 10 May 2023

Different Types of Services



Source: Adapted from Trott, P. (2017). *Innovation Management and New Product Development*. 6th edition. Pearson.

Characteristics of Services

Services as a process are co-produced by the consumer.

Operations staff

- Play both an operation and marketing role when dealing with the consumer.

Intangibility

- Services cannot be protected by patents and are easily copied. Often this results in “me too” services.

Heterogeneity

- Lack of uniformity results in difficulties developing accurate concept description and can create problems in concept testing.

Simultaneous production and consumption

- Services are produced and consumed simultaneously.

Perishability

- Services cannot be saved, stored, or returned.

Source: Adapted from Trott, P. (2017). *Innovation Management and New Product Development*. 6th edition, Pearson.

Classification of Services

Classification	Description	Example
Major Innovation	Computer-based technologies & data drive for new & undefined markets.	ChatGPT 
Start-up business	New services introduced into the market that is already served by other existing services.	42Race.com 
New services for the market presently served	New service offerings to existing customers of a company (although the services may be available from other companies).	Nanyang Old Coffee 
Service line extensions	Expansion to the existing service line, such as adding new menu items, new routes and new courses.	ComfortDelgro 
Service improvements	Features changes in the services that are currently being offered.	Singapore Airlines 
Style changes	Aesthetic or visual changes to impact customer perceptions without any fundamental changes in service.	Singapore Police Force Crest 

Source: Adapted from Trott, P. (2017). *Innovation Management and New Product Development*. 6th edition. Pearson.

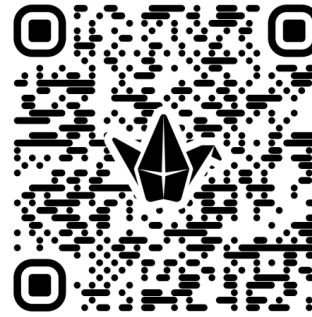
Discussion 5.1

‘The everything app’: why Elon Musk wants X to be a WeChat for the west

Twitter owner’s rebranding is part of larger strategy to add payment functionality and more



📷 WeChat is used for messaging, email, social media, payments, food deliveries, plane and train tickets, ride hailing and much more. Photograph: Omar Marques/Sopa Images/Shutterstock

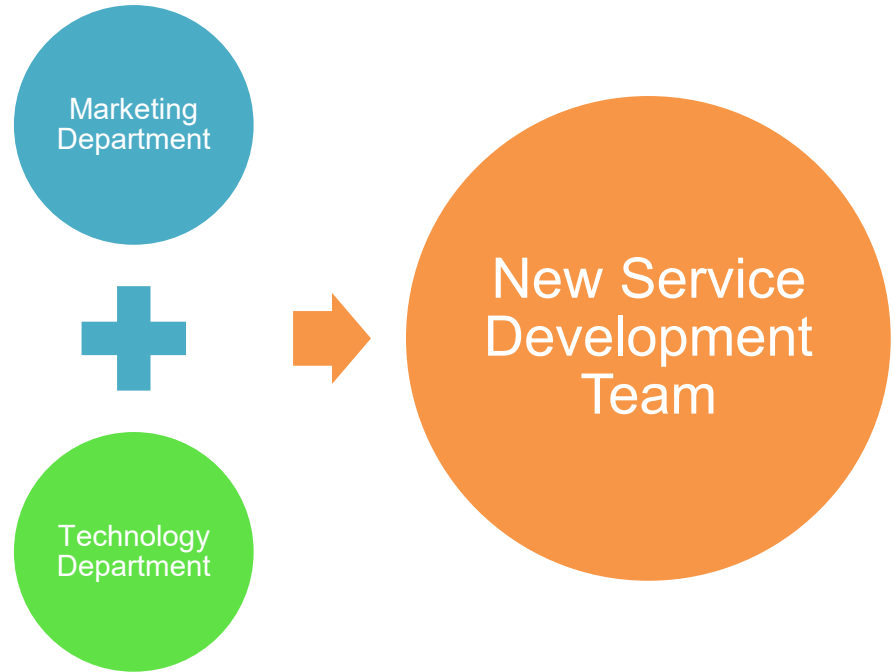


Read: Milmo, D., Hawkins, A. (2023). *“The everything app’: why Elon Musk wants X to be a WeChat for the west.* The Guardian. Retrieved from <https://www.theguardian.com/media/2023/jul/29/elon-musk-wechat-twitter-rebranding-everything-app-for-west> on 1 Aug 2023

1. What classification of service would this “X” (formerly, Twitter) innovation fall under?
2. Would it be possible for “X” to be like “WeChat? Why or Why not?

Technology and New Service Development

Technology has become the most significant enabler of innovation in services.



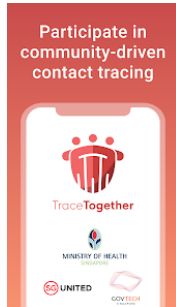
New Service Development Models – Iterative Model



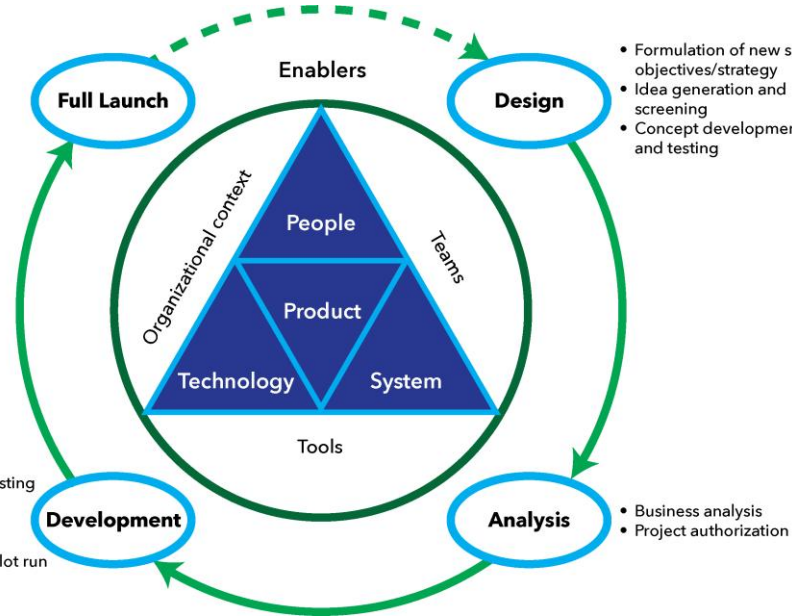
4 stages:
Design
Analysis
Development
Full Launch



Source:
<https://www.todayonline.com/singapore/contact-tracing-data-pandemic-era-tracetogether-safeentry-systems-deleted-2356446>



- Full-scale Launch
- Post-Launch Review



A Critical Evaluation of the New Service Development Process: Integrating Service Innovation and Service Design. Sage Publications. Source: Adapted from Johnson, S., Menor, L.J., Roth, A., Chase, R.B..(2000).

Classification of PSI

Classification	Examples
New Service Innovation	Polytechnic Preparation Programme for good ITE students
Process Innovation	a. Technology Process : Online Road Tax Payment via OneMotoring website b. Administrative Process: Health Buddy app for medical and dental appointments
Policy Innovation	Research, Innovation and Enterprise Plan by National Research Foundation
Social Innovation	Blue Recycling Bin by National Environmental Agency
Innovation in Governance	GeBiz for public tenders
Conceptual Innovation	MOH Healthier SG for seniors with their regular doctors.

Collaboration in PSI



Open Innovation and a System Approach

OPEN 
INNOVATION
NETWORK

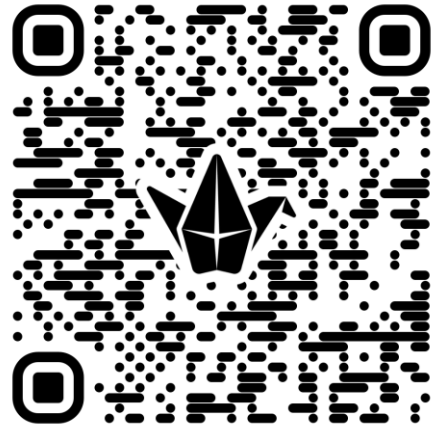
<https://www.openinnovationnetwork.gov.sg/>

Source: Adapted from Trott, P. (2021). *Innovation Management and New Product Development*. 7th edition. Pearson.

Discussion 5.2

(Do as an individual activity and submit it onto Padlet with your name)

- Goto <https://www.openinnovationnetwork.gov.sg/> and share interesting discovery of:
 1. Past Unique Innovation Solutions
 2. Interesting Upcoming Innovation Challenges



Governance Paradigms and PSI

	Traditional Public Administration	New Public Management	New Public Governance
Emphasis	Rule of Law and standardisation	Efficiency through adoption of private sector practices	Collective action to solve complex policy problems
Innovation agenda	New Regulations	Efficiency and Accountability	Citizen Participation
Entrepreneurs-Initiators	Policy Entrepreneurs	Managers and Frontline Staff as Entrepreneurs	Social Entrepreneurs
The role of Public Organisations	Implementers of mandated change	Source of innovation via Public Entrepreneurs	Meta-governors of Innovation Process
The role of Citizens	Holders of legal rights and passive users of innovations	Rational customers towards innovations	Co-innovators in the centre of innovation
The role of Businesses	Pressure groups	Privatization stakeholders	Partners in co-creation of solutions

Source: Trott, P. (2021). *Innovation Management and New Product Development*. 7th edition. Pearson.

Future Service Trends

1. Personalisation

- Customised approach for each consumer to build relationship

2. Understanding Customer Expectation

- Service must meet consumer expectations to create value

3. Credibility

- Deliver the emotional aspect of the brand promise

4. Simplification

- Interactive and engaging info-mediaries
- Services are consumed in easy, comfortable and convenient way.

5. Internationalisation

- Available on a global scale.

6. Bundling

- Combination of product and service are delivered.

7. Market Spaces

- More than one services are offered.

Source: Retrieved from <https://www.mbaknol.com/services-marketing/10-trends-identified-in-the-service-sector-in-21st-century/> on 20 July 2023.

Recommended Readings

- Milmo, D., Hawkins, A. (2023). *“The everything app”: why Elon Musk wants X to be a WeChat for the west*. The Guardian. Retrieved from <https://www.theguardian.com/media/2023/jul/29/elon-musk-wechat-twitter-rebranding-everything-app-for-west> on 1 Aug 2023
- Just In Mind (n.d.) <https://www.justinmind.com/blog/mobile-app-wireframe> on 12 Jan 2022

End

- Preparation & Consultation Time for Presentation